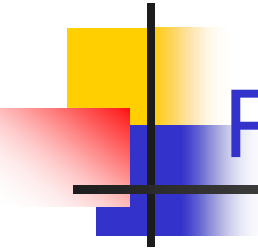


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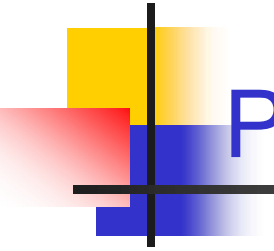
Installation & Administration of Network Services

R. NOUARA



Package Management.

- Ubuntu use Advanced Package Tool (APT) command to manage packages. As all Debian-based Linux distributions.
- **Package Management:** allows users to easily **install**, **update** or **remove** software packages.
- **Dependency Resolution:** identify and install automatically the depending packages.
- **Repository Management:** repositories are centralized locations where packages are stored.
- **System Updates:** facilitates keeping the system up-to-date.
- **User-Friendly Interface:** a simplified and more interactive command-line interface



Package Management.

- To view all the available packages on Ubuntu system:

`apt list`

- packages that have been installed.

`apt list --installed`

- determine if the package is available in the Ubuntu repository

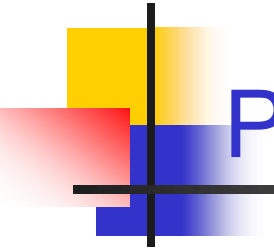
`apt search ftp`

- Displaying the details of a package

`apt show nano`

- To examine the dependencies of a package

`apt depends vim`



Package Management with apt.

- To check the installed package version

```
apt-cache policy nano
```

```
apt install --just-print nano
```

- install a package

```
apt install nano -y
```

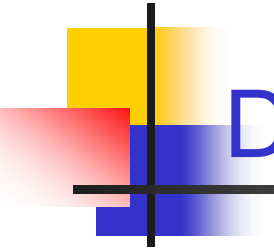
- install a specific version of a package

```
apt install <package_name>=<version_number>
```

```
apt install mysql-server=8.0.28-0ubuntu4
```

- to uninstall a package:

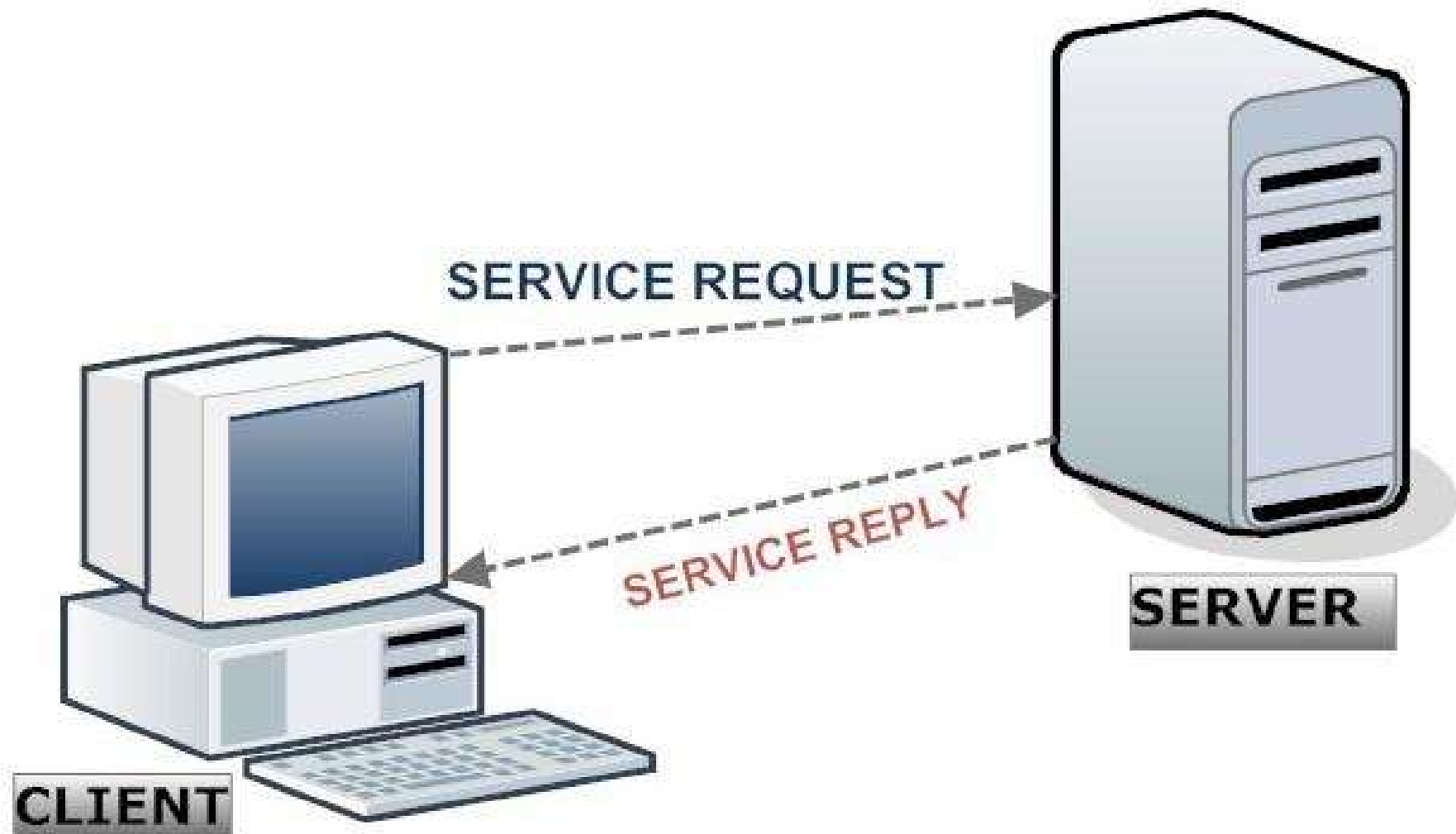
```
apt remove nano -y
```

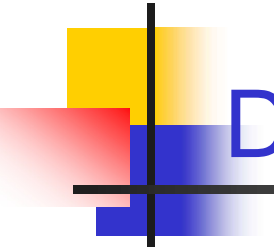


Definition of a service.

- Network Service corresponds to one or more processes (**daemon**) that run on a server in the background.
- Responsible for performing system tasks **to provide services to clients** (example: Databases).
- Accessible to clients (**users**) on the network according to predefined access rules (**protocol**).
- ***A port number is assigned to a each service.***

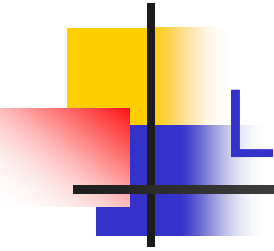
Definition of a service.





Definition of a service.

- **LDAP Directory service**
- **Email server**
- **File sharing**
- **Instant messaging services**
- **Printing service**
- **Online gaming**
- **File sharing.....**

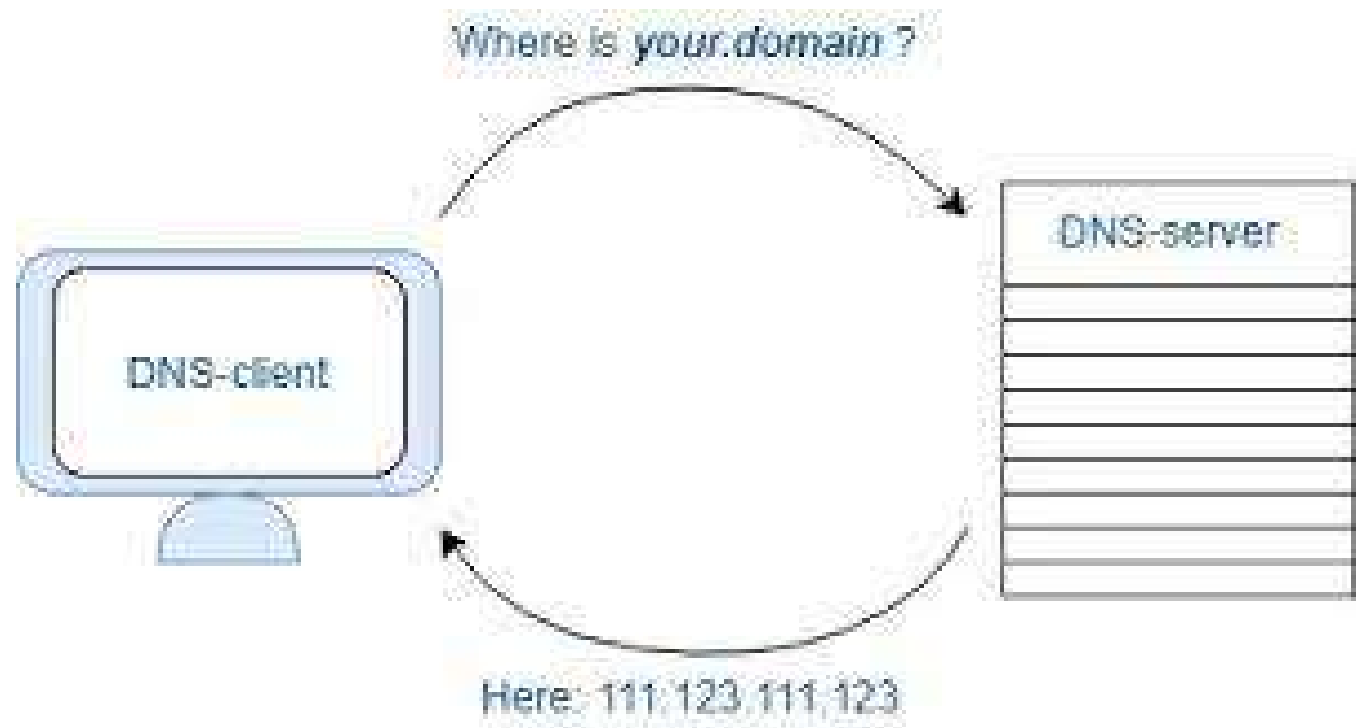


Life cycle of a service

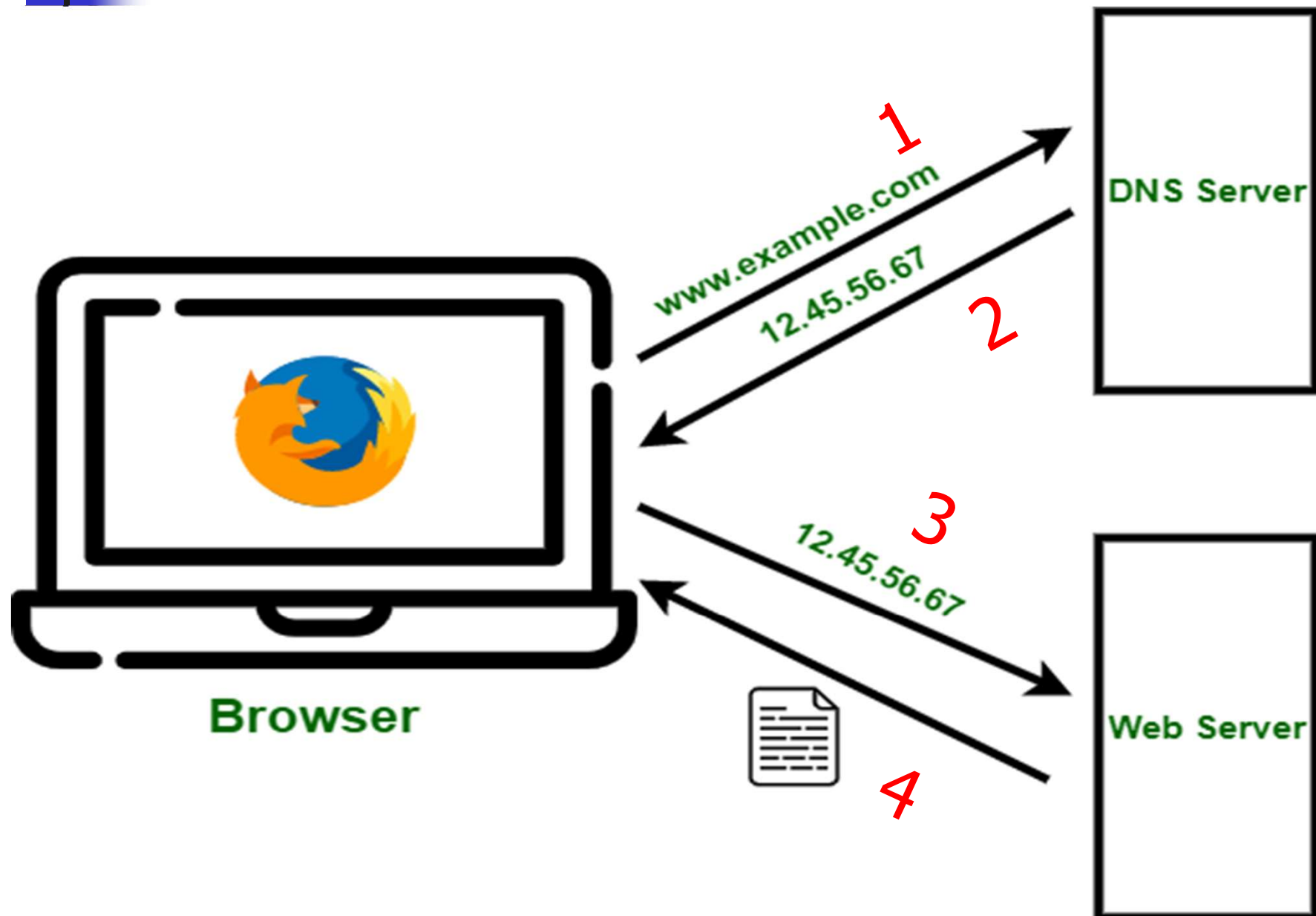
- **Installation** : in this step we install the service with apt command.
- **Configuration**: in thi we configure the service to specificities of the enterprise.
- **Administration**: This step is the long one, in which we exploit the service and allow users to access.

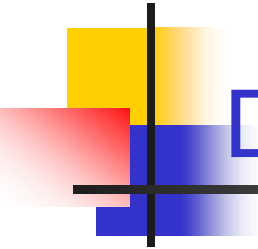
Example of service

- Example :
- **DNS server:** gives the **ip** address of a corresponding domaine name.
- Port : **53**



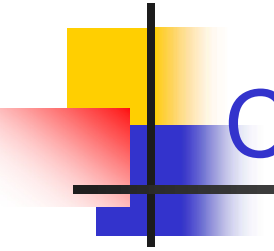
Example of service: DNS





DNS server.

```
tm1 [En fonction] - Oracle VM VirtualBox  
[root@localhost ~]# nslookup univ-constantine2.dz  
Server:                8.8.8.8  
Address:                8.8.8.8#53  
  
Non-authoritative answer:  
Name:   univ-constantine2.dz  
Address: 144.91.73.190  
  
[root@localhost ~]# _
```



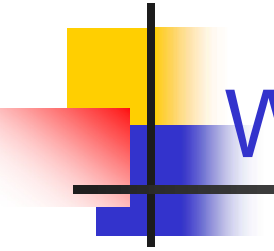
Command to manage service

Systemctl <option> service

Option can be:

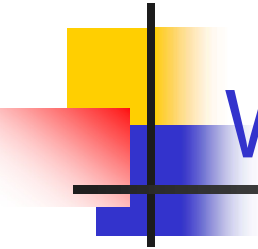
- **status** : to display the status of a service
- **start** : to run the service
- **stop** : to stop running the service
- **restart**: stop and start the service
- **enable/disable**: to enable or disable start of the service when booting the system.

Systemctl start apache2



Web server Apache.

- Example :
- Web Server ***Apache***.
- Nom de service: **apache2**
- Port : **80**
- To install
`apt install apache2`

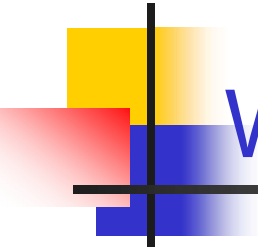


Web server Apache.

- If you have error to install package:
- **Error:failed to fetch..**

```
W: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/precise/InRelease
W: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/precise-updates/InRelease
W: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/precise-backports/InRelease
W: Failed to fetch http://security.ubuntu.com/ubuntu/dists/precise-security/InRelease
W: Failed to fetch http://extras.ubuntu.com/ubuntu/dists/precise/InRelease
W: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/precise/Release.gpg
W: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/precise-updates/Release.gpg
```

- Run the command :
apt update



Web server Apache

- Configuration :

`/etc/apache2/apache2.conf`

- The configuration file of the service.

- Listen ports

`/etc/apache2/ports.conf`

- Website(s)

- **`/etc/apache2/sites-available/`**